SPA617: Radio Astronomy Laboratory

Laboratory Report

RF Transmission, Reflections, and Standing Waves

Name: Bhawna Parashar

Roll No. : 252310602

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1. Title

Experimental Study of RF Transmission Lines, Signal Reflections, Standing Waves, and the Electrical Properties of a Coaxial Cable.

2. Objective

The main objective of this experiment was to understand how radio-frequency signals propagate through a coaxial cable and how reflections occur when the cable is not properly terminated.

The specific objectives were:

1. To determine the phase velocity and velocity factor of a signal travelling through a coaxial cable by measuring the frequency spacing between consecutive maxima or minima.
2. To study signal reflections from an open-ended cable and a cable terminated with a $50\,\Omega$ load.
3. To estimate the reflection coefficient and examine whether the cable is approximately matched to a $50\,\Omega$ termination.
4. To calculate the inductance and capacitance per unit length of the cable using the measured phase velocity and the approximate characteristic impedance.
5. To measure the insertion loss and estimate the attenuation per unit length of the coaxial cable.

3. Apparatus Used

The following instruments and components were used in the experiment:

- RF signal generator
- Digital storage oscilloscope (DSO)
- Spectrum analyser
- RF Tee connector
- Coaxial cable under test
- Short reference coaxial cable
- $50\,\Omega$ termination
- Open-circuit termination
- Suitable RF adapters and connectors

4. Theory

4.1 Transmission Lines and Signal Propagation

A transmission line is used to carry electrical signals from one point to another. Coaxial cables are widely used in RF systems because they guide electromagnetic energy while reducing unwanted radiation and external interference.

A coaxial cable consists of an inner conductor surrounded by a dielectric material and an outer conductor. The cable supports transverse electromagnetic (TEM) propagation, where the electric and magnetic fields are perpendicular to the direction of propagation.

The signal travels more slowly through the cable than through a vacuum because of the dielectric material between the conductors.

The behaviour of a transmission line depends on its characteristic impedance, the load connected to its end, and the frequency of the applied signal.

4.2 Signal Reflection and Standing Waves

When a travelling wave reaches the end of a cable, the amount of signal reflected depends on the load impedance Z_L and the characteristic impedance Z_0 of the cable.

If these impedances are different, a part of the incident signal is reflected towards the source. The incident and reflected waves superimpose, producing constructive or destructive interference.

The voltage reflection coefficient is

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}. \tag{1}$$

Three important cases are:

- **Open circuit:** For an ideal open circuit, $Z_L \to \infty$, and $\Gamma = +1$. The voltage reflection is complete and has the same polarity as the incident wave.
- **Short circuit:** For an ideal short circuit, $Z_L = 0$, and $\Gamma = -1$. The reflected voltage has opposite polarity.
- **Matched load:** When $Z_L = Z_0$, $\Gamma = 0$. Ideally, no signal is reflected.

In this experiment, the oscilloscope measures the voltage at a fixed point, namely the Tee junction. It does not directly display the spatial standing-wave pattern along the cable.

When the signal frequency is varied, the phase of the reflected signal changes. This causes the measured voltage amplitude at the Tee junction to vary periodically.

The maximum and minimum voltage amplitudes are given by

$$V_{\max} = |V_{\text{inc}}| + |V_{\text{ref}}|, \tag{2}$$

and

$$V_{\min} = |V_{\text{inc}}| - |V_{\text{ref}}|, \tag{3}$$

assuming that the incident-wave amplitude is at least as large as the reflected-wave amplitude.

Combining these expressions gives the magnitude of the reflection coefficient:

$$|\Gamma| = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}}. \tag{4}$$

This expression allows the reflection coefficient to be estimated from the measured voltage maxima and minima.

4.3 Phase Velocity and Velocity Factor

The phase velocity v_p is the speed at which a particular phase of the RF signal travels through the cable.

Let the physical cable length be ℓ. When the signal reflects from the far end, it travels a total distance 2ℓ during a round trip.

The round-trip time delay is therefore

$$\tau = \frac{2\ell}{v_p}. \tag{5}$$

The round-trip phase changes with frequency. Two consecutive maxima, or two consecutive minima, occur when the frequency changes by an amount Δf that produces a phase change of 2π.

Therefore,

$$2\pi\Delta f\,\tau = 2\pi. \tag{6}$$

Simplifying,

$$\tau = \frac{1}{\Delta f}. \tag{7}$$

Substituting the round-trip delay gives

$$\boxed{v_p = 2\ell\Delta f}. \tag{8}$$

Here, Δf is the frequency difference between two consecutive maxima or two consecutive minima, expressed in hertz.

The velocity factor is the ratio of the phase velocity inside the cable to the speed of light in vacuum:

$$\boxed{VF = \frac{v_p}{c}}, \tag{9}$$

where

$$c \approx 3.0 \times 10^8\,\mathrm{m\,s^{-1}}. \tag{10}$$

The velocity factor is dimensionless and is generally less than one.

4.4 Characteristic Impedance

The characteristic impedance Z_0 describes the ratio of voltage to current for a travelling wave on a transmission line. It depends on the cable's distributed inductance and capacitance.

For a low-loss transmission line,

$$Z_0 = \sqrt{\frac{L'}{C'}}, \tag{11}$$

where L' is the inductance per unit length and C' is the capacitance per unit length.

In this experiment, a $50\,\Omega$ termination is connected to the far end of the cable. The reflection pattern is compared with the open-circuit condition.

A substantial reduction in the amplitude variation when the $50\,\Omega$ load is connected indicates that the cable is approximately matched to this termination.

This comparison provides an approximate verification of the cable's characteristic impedance.

4.5 Inductance and Capacitance per Unit Length

For a low-loss transmission line, the phase velocity is

$$v_p = \frac{1}{\sqrt{L'C'}}. \tag{12}$$

The characteristic impedance is

$$Z_0 = \sqrt{\frac{L'}{C'}}. \tag{13}$$

Rearranging these equations gives the inductance per unit length:

$$\boxed{L' = \frac{Z_0}{v_p}}. \tag{14}$$

Similarly, the capacitance per unit length is

$$\boxed{C' = \frac{1}{Z_0 v_p}}. \tag{15}$$

The units are H/m for inductance per unit length and F/m for capacitance per unit length.

These quantities describe the distributed electrical properties of the cable.

4.6 Cable Insertion Loss and Attenuation

As an RF signal travels through a real cable, some of its power is lost because of conductor resistance, dielectric losses, and other non-ideal effects.

The insertion loss is determined by comparing the power measured using a short reference connection with the power measured after inserting the cable under test.

It is calculated as

$$\boxed{IL = P_{\text{ref}} - P_{\text{cable}}}, \tag{16}$$

where both power values are expressed in dBm. The resulting insertion loss is expressed in dB.

The attenuation per unit length is

$$\boxed{\alpha = \frac{IL}{\ell} \text{ dB/m}}. \tag{17}$$

A larger attenuation value indicates greater signal loss per unit length of cable.

5. Procedure

5.1 Part A: Measurement of Phase Velocity

1. The physical length of the coaxial cable was measured and recorded.
2. The signal generator, Tee connector, cable under test, and oscilloscope were connected according to the experimental setup in the laboratory manual.
3. The far end of the cable was left open.
4. The oscilloscope input impedance was set to $1\,\text{M}\Omega$ to avoid loading the Tee junction with a $50\,\Omega$ input.
5. A sinusoidal signal of constant amplitude was applied to the cable.
6. The signal frequency was varied slowly while observing the voltage amplitude on the oscilloscope.
7. The frequencies corresponding to two consecutive maxima or two consecutive minima were recorded.
8. The frequency difference Δf was calculated, and the phase velocity and velocity factor were obtained using the equations given in the theory section.

5.2 Part B: Reflection Coefficient

1. The cable was kept open at its far end, and the maximum and minimum voltage amplitudes were recorded over one ripple period.
2. The reflection coefficient for the open-ended cable was calculated using

$$|\Gamma_{\text{open}}| = \frac{V_{\text{max,open}} - V_{\text{min,open}}}{V_{\text{max,open}} + V_{\text{min,open}}}.$$

3. A $50\,\Omega$ termination was connected to the far end of the cable.
4. The frequency sweep was repeated without changing the signal-generator amplitude or the other instrument settings.
5. The maximum and minimum voltages were recorded for the terminated cable.
6. The reflection coefficient was calculated using

$$|\Gamma_{50}| = \frac{V_{\text{max,50}} - V_{\text{min,50}}}{V_{\text{max,50}} + V_{\text{min,50}}}.$$

7. The two reflection coefficients were compared to assess the matching between the cable and the $50\,\Omega$ termination.

5.3 Part C: Inductance and Capacitance

1. The phase velocity obtained in Part A was used.
2. The characteristic impedance was taken to be approximately $50\,\Omega$, based on the termination comparison.
3. The inductance and capacitance per unit length were calculated using

$$L' = \frac{Z_0}{v_p}, \qquad C' = \frac{1}{Z_0 v_p}.$$

4. The results were expressed in H/m and F/m, respectively, with suitable conversions to nH/m and pF/m if required.

5.4 Part D: Measurement of Cable Attenuation

1. The signal generator was connected to the spectrum analyser using a short reference cable and suitable adapters.
2. A suitable test frequency and input power were selected and kept constant.

3. The reference power P_{ref} was recorded in dBm.

4. The short reference connection was replaced by the cable under test, keeping the remaining setup unchanged.

5. The output power P_{cable} was recorded.

6. The insertion loss was calculated by subtracting the cable power from the reference power.

7. The attenuation per unit length was calculated by dividing the insertion loss by the physical cable length.

6. Reading and Observation Tables

Part A: RF – Transmission, Reflections and Standing Waves

6.1 Table 1: Phase Velocity Measurement for open ended cable

S.No.	Frequency (MHz)	RMS	Amplitude	Remark
1	20	890 mV	1.99 mV	
2	20	1.57 V	4.5 mV	
3	25	1.5 V	4 V	
4	58	67 mV	240 mV	min
5	68	630 mV	1.76 V	
6	78	1.2 V	3.36 V	
7	88	1.56 V	4.4 V	
8	98	1.73 V	4.8 V	
9	108	1.74 V	4.84 V	max
10	118	1.64 V	4.6 V	
11	128	1.47 V	4.12 V	
12	138	1.21 V	3.36 V	
13	148	920 mV	2.6 V	
14	158	640 mV	1.8 V	
15	168	284 mV	880 mV	
16	173	97 mV	320 mV	min
17	178	360 mV	1.12 V	
18	198	1.28 V	3.6 V	
19	210	1.41 V	3.96 V	max
20	218	1.28 V	3.6 V	
21	238	938 mV	2.6 V	

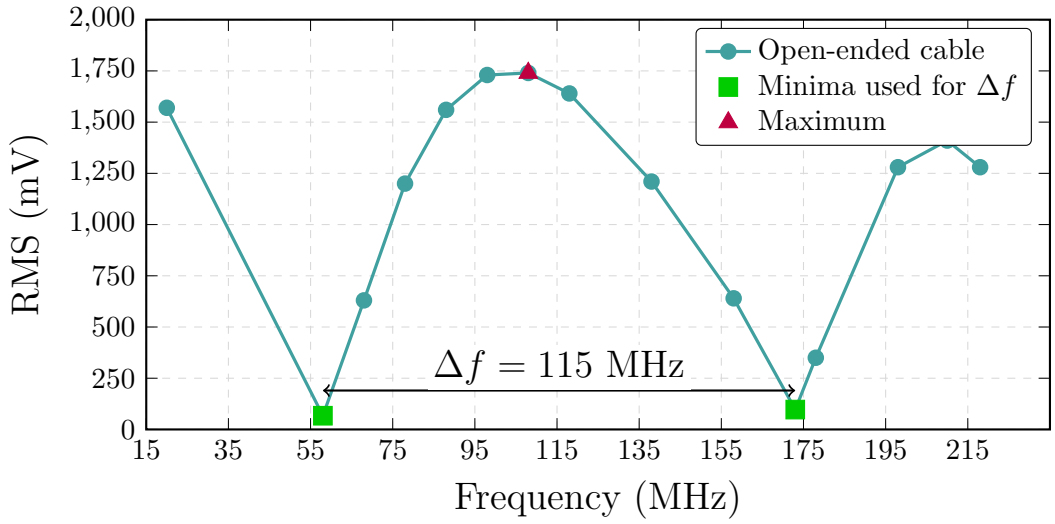


Figure 1: RMS voltage at the Tee junction versus frequency for the open-ended cable.

Cable length: 92 cm

First maximum frequency:108 MHz

Next maximum frequency:210 MHz

Frequency spacing: Δf = 210-108=102 MHz=102000000 Hz

$$since \boxed{v_p = 2\ell\Delta f}. \tag{18}$$

$v_p = 2 * 0.92 * 102000000 = 187680000$ **m/s**

$$\boxed{VF = \frac{v_p}{c} = \frac{187680000}{300000000} = \mathbf{0.6256}}, \tag{19}$$

Part B: Load Connected = 50 Ω

6.2 Table 2: Reflection Coefficient

$$\Gamma_{\text{open}}| = \frac{1.74 - 0.097}{1.74 + 0.097} = \frac{1.643}{1.837} = 0.89439.$$

$$|\Gamma_{50}| = \frac{0.990 - 0.9}{0.990 + 0.9} = \frac{0.090}{1.89} = 0.04761.$$

S.No.	Frequency (MHz)	RMS (mV)	Amplitude (V)	Remark
1	20	973.26	2.8	
2	20	–	–	
3	25	1040	2.8	max
4	58	954.19	2.68	
5	68	982.07	2.68	
6	78	972.11	2.72	
7	88	973.04	2.72	
8	98	990.06	2.72	max
9	108	982	2.72	
10	118	950	2.68	
11	128	942	2.6	
12	138	926	2.56	
13	148	900	2.48	min
14	158	904	2.52	
15	168	933	2.56	
16	173	938	2.64	
17	178	950	2.64	max
18	198	929	2.56	
19	210	812	2.24	
20	218	731	2.04	
21	238	691	1.88	

Table 1: Comparison of voltage amplitudes for open-circuit and $50\,\Omega$ termination conditions.

Termination	$V_{\max}$ **(V)**	$V_{\min}$ **(V)**	$\lvert\Gamma\rvert$
Open circuit	1.74	0.097	0.89439
$50\,\Omega$ load	0.990	0.900	0.04761

6.3 Inductance and Capacitance per Unit Length

Taking $Z_0 \approx 50\,\Omega$,

$$L' = \frac{50}{v_p}\ \text{H/m} = \frac{50}{187680000} = 2.66 \times 10^{-7}\ \text{H/m}.$$

Similarly,

$$C' = \frac{1}{50 v_p}\ \text{F/m} = \frac{1}{50 \times 187680000} = \frac{1}{9384000000} = 1.06 \times 10^{-10}\ \text{F/m}.$$

6.4 Cable Attenuation

S.No.	Frequency (MHz)	P_{ref} (dBm)	P_{cable} (dBm)	I.L.(dB)	α(dB/m)
1	50	0	-1.92	$+1.92$ dBm	2.086

7. Results and Discussion

Table 2: Summary of measured and calculated transmission-line parameters.

Quantity	**Symbol**	**Measured/Calculated Value**
Cable length (m)	ℓ	$0.92\,\text{m}$
Frequency spacing (Hz)	Δf	$102\,\text{MHz}$
Phase velocity (m/s)	v_p	$1.8768 \times 10^8\,\text{m/s}$
Velocity factor	v_p/c	0.6256
Approximate characteristic impedance (Ω)	Z_0	$50\,\Omega$
Inductance per unit length (H/m)	L'	$2.66 \times 10^{-7}\,\text{H/m}$
Capacitance per unit length (F/m)	C'	$1.06 \times 10^{-10}\,\text{F/m}$
Attenuation (dB/m)	α	$2.086\,\text{dB/m}$

The measured and calculated parameters provide an understanding of the electrical characteristics of the transmission line. The cable length was measured as $0.92\,\text{m}$, and the frequency spacing was determined to be $102\,\text{MHz}$. The phase velocity was calculated to be $1.8768 \times 10^8\,\text{m/s}$, corresponding to a velocity factor of approximately 0.6256. Thus, the electromagnetic signal propagates through the cable at about 62.56% of the speed of light in vacuum.

The approximate characteristic impedance was found to be $50\,\Omega$, consistent with the nominal impedance of commonly used RF coaxial cables. The inductance and capacitance per unit length were calculated as $2.66 \times 10^{-7}\,\text{H/m}$ and $1.06 \times 10^{-10}\,\text{F/m}$, respectively. These parameters describe the energy-storage properties of the transmission line.

The attenuation was determined to be $2.086\,\text{dB/m}$. It represents the signal loss per unit length due to conductor losses, dielectric losses, and other frequency-dependent effects. Overall, the experiment demonstrates how transmission-line measurements can be used to estimate propagation velocity, characteristic impedance, inductance, capacitance, and attenuation. Small deviations from ideal values may arise from measurement uncertainties and experimental conditions.

8. The Part I Liked the Most

The part I found most interesting was observing how the signal amplitude changed when the frequency of the generator was varied. Initially, the change in amplitude might appear unexpected, because the signal-generator amplitude was kept constant. However, the experiment showed that the incident and reflected waves interfere with each other, producing maxima and minima at different frequencies.

I also found the comparison between the open-ended cable and the cable connected to a $50\,\Omega$ termination useful. It demonstrated how impedance matching reduces reflections and improves signal transmission.

This experiment helped me connect the theory of electromagnetic-wave propagation with practical RF measurements.

9. Key Takeaways

- A coaxial cable guides electromagnetic signals from one point to another.
- A mismatch between the cable impedance and load impedance causes part of the incident signal to reflect.
- The superposition of incident and reflected waves produces constructive and destructive interference.
- The frequency spacing between consecutive amplitude maxima or minima can be used to determine the phase velocity and velocity factor.
- A $50\,\Omega$ termination reduces reflections when the cable is approximately matched to $50\,\Omega$.
- The measured phase velocity and characteristic impedance can be used to estimate the inductance and capacitance per unit length.
- Comparing reference power with cable output power provides an estimate of insertion loss and attenuation per unit length.
- Consistent instrument settings, suitable impedance termination, and unchanged cable connections are important for reliable RF measurements.

Overall, this experiment provided a practical understanding of signal propagation, impedance matching, standing-wave formation, and attenuation in coaxial transmission lines. These concepts are important in RF engineering and radio astronomy, where weak signals must be transmitted and measured with minimal loss and unwanted reflections.